

SACHI KAUSHIK

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EDUCATION

Columbia University

M.S. in Computer Science, GPA: 3.5/4.0

New York, NY

Exp Dec 2025

Vellore Institute of Technology

B.Tech. in Computer Science and Engineering, GPA: 9.32/10.0

Vellore, IN

May 2024

WORK EXPERIENCE

adMarketplace

New York, NY

Machine Learning Engineer Intern

Jun 2025 – Present

- Engineering an ad ranking system that integrates transformer embeddings, FAISS vector stores with 10M+ entries, and real-time streaming pipelines to generate CTR predictions at scale.
- Improved a time series CTR prediction model by incorporating demographic features like age, achieving a 24% reduction in RMSE and demonstrating measurable gains in personalization effectiveness.
- Modeled DistilGPT-2 on partial-to-complete query term pairs for real-world query expansion, increasing correct expansions by 17% and enabling faster, more relevant ad matches from incomplete inputs.

Columbia Build Lab

New York, NY

Software Engineer

Jan 2025 – May 2025

- Architected 2 end-to-end modules for a dentist referral platform by integrating Flask APIs with React, built 12 reusable components, reducing future development effort by 30% and enabling downstream analytics.
- Facilitated cross-functional collaboration with a designer and developer in a lean startup environment to define scope, refine data-driven UI/UX, and deliver the MVP in 10 sprints.

Tech Mahindra Maker's Lab Innovation Cell

Pune, IN

Software Development Intern

May 2023 – Aug 2023

- Designed and deployed a Knowledge Management portal chatbot as an automation tool for Tech Mahindra's website, improving information access design patterns and reducing response times by 40%.
- Integrated a next-word recommendation system leveraging language modeling into the portal (100k+ users), reducing average typing length per query by 12%.

TECHNICAL SKILLS

Coursework: Analysis of Algorithms, Applied Machine Learning, Cloud Computing, Advanced Database Systems

Programming Languages: C, C++, Python, R, Java, HTML5, CSS, JavaScript, SQL, LaTeX, Go, TypeScript, Scala

Libraries & Frameworks: Spring, MapReduce, pandas, matplotlib, jQuery, GraphQL, Flask, Vue, Angular, Spark

Software & Tools: MySQL, Git, Node.js, Azure, GCP, AWS, Kafka, Docker, Linux, Agile, Kubernetes, Jenkins

PROJECTS

Improving Information Retrieval using User Feedback

Jan 2025 – May 2025

- Developed a Python-based query reformulation system with Google Custom Search API, iteratively expanding and reordering queries with user feedback, supporting 50+ sessions per run.
- Achieved ≥ 0.9 precision@10 for ambiguous queries by appending up to 2 discriminative terms per iteration, reducing irrelevant top-10 results by 70%.

Cloud Casino – Cloud-Based Microservice Casino Platform

Sep 2024 – Dec 2024

- Collaborated on a scalable casino web application built with AWS and FastAPI, featuring Google OAuth login, game stat tracking, and real-time analytics with < 200 ms API latency across multiple microservices.
- Deployed 5 microservices across AWS EC2 and Google PaaS, configured API Gateway with JWT authentication blocking 100% unauthorized requests, and automated signup notifications via SES + Lambda.

Safe-Web for Information Security Management

Sep 2023 – Dec 2023

- Addressed 4 critical security issues (image-based passwords, keyloggers, hashing, XSS) and implemented 3 protections (cookie security, picture encryption, Solidity), ensuring compliance with OWASP Top-10 standards.
- Fashioned an interactive, user-friendly React.js website covering 10+ cybersecurity threats and 20+ best practices to educate users on secure habits and awareness.